
CIS437 - Cloud Computing

– Google Cloud Setup & First VM –

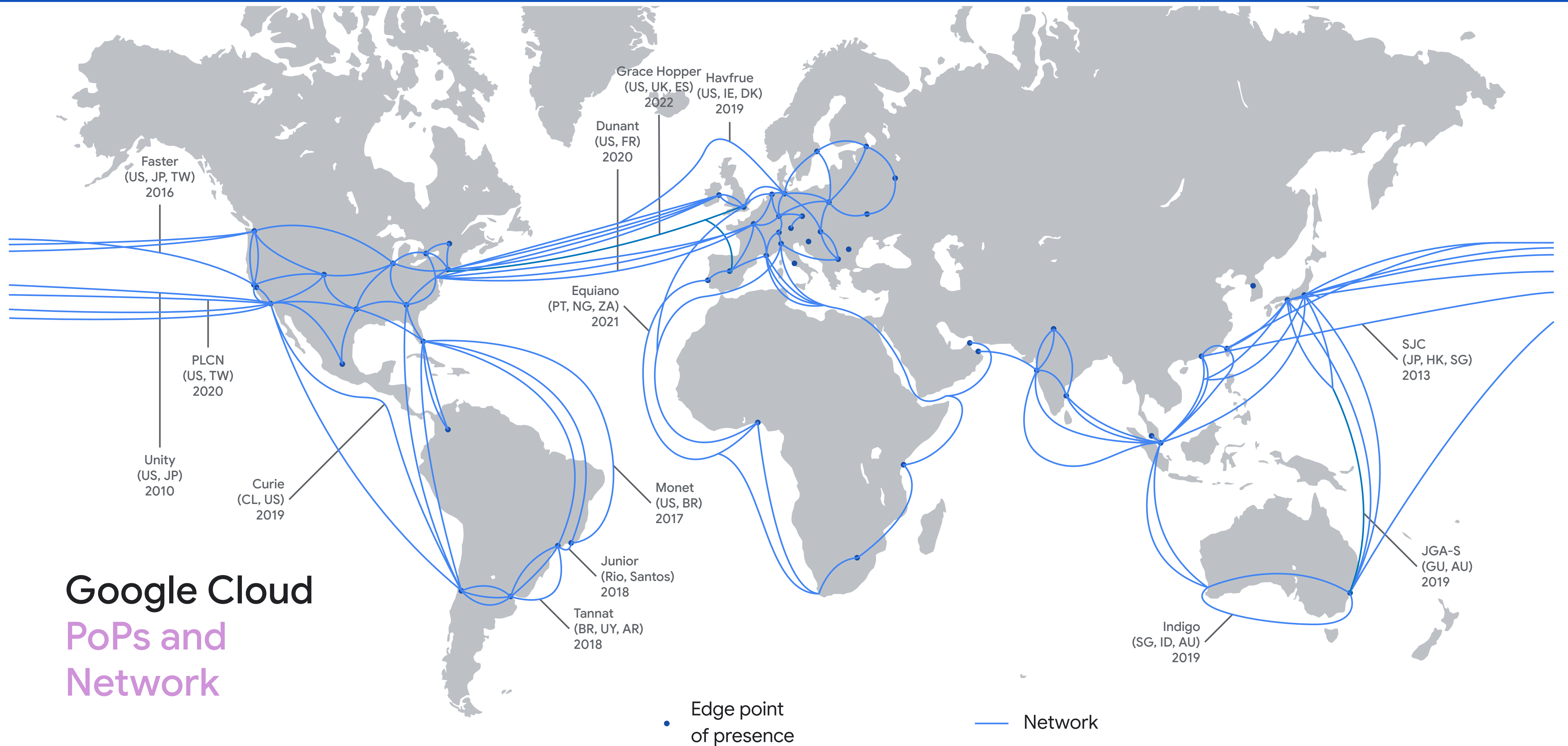
— Instructor: Yong Zhuang —

yong.zhuang@gvsu.edu

Agenda

- Google Cloud Console
- Credits
- Projects
- Virtual Machines
- Assignment 1

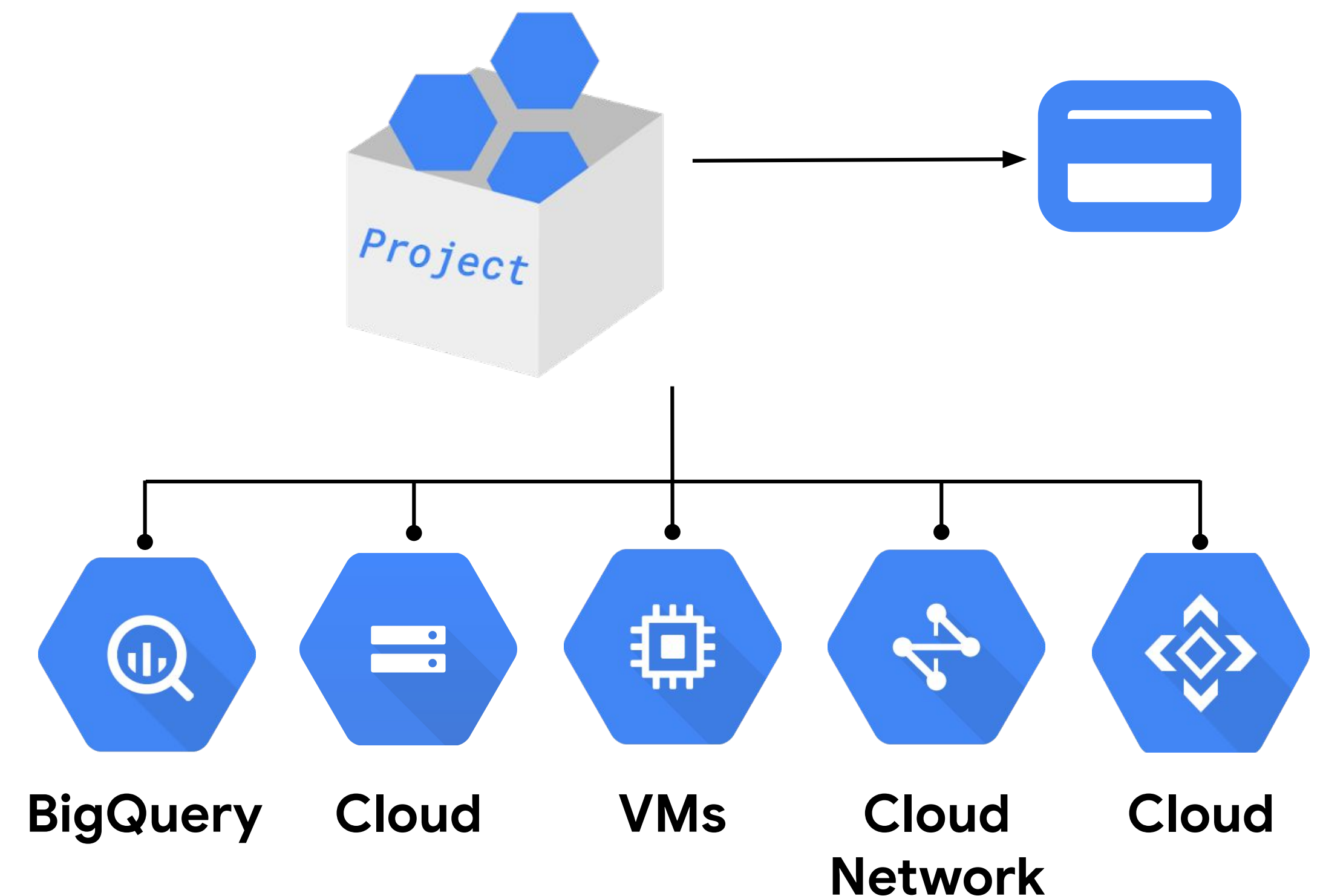
Google Cloud PoPs and Network



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How billing works

- ✓ Billing account pays for project resources.
- ✓ A billing account is linked to zero or more projects.
- ✓ Accounts are charged automatically, invoiced monthly, or invoiced at the threshold limit.
- ✓ Sub accounts can be used for separate billing for projects.



Budgets and alerts keep your billing under control

Google Cloud Platform

Project

Billing

Overview

Budgets & alerts

Transactions

Billing export

Payment settings

Budgets & alerts

Corporate Billing Account

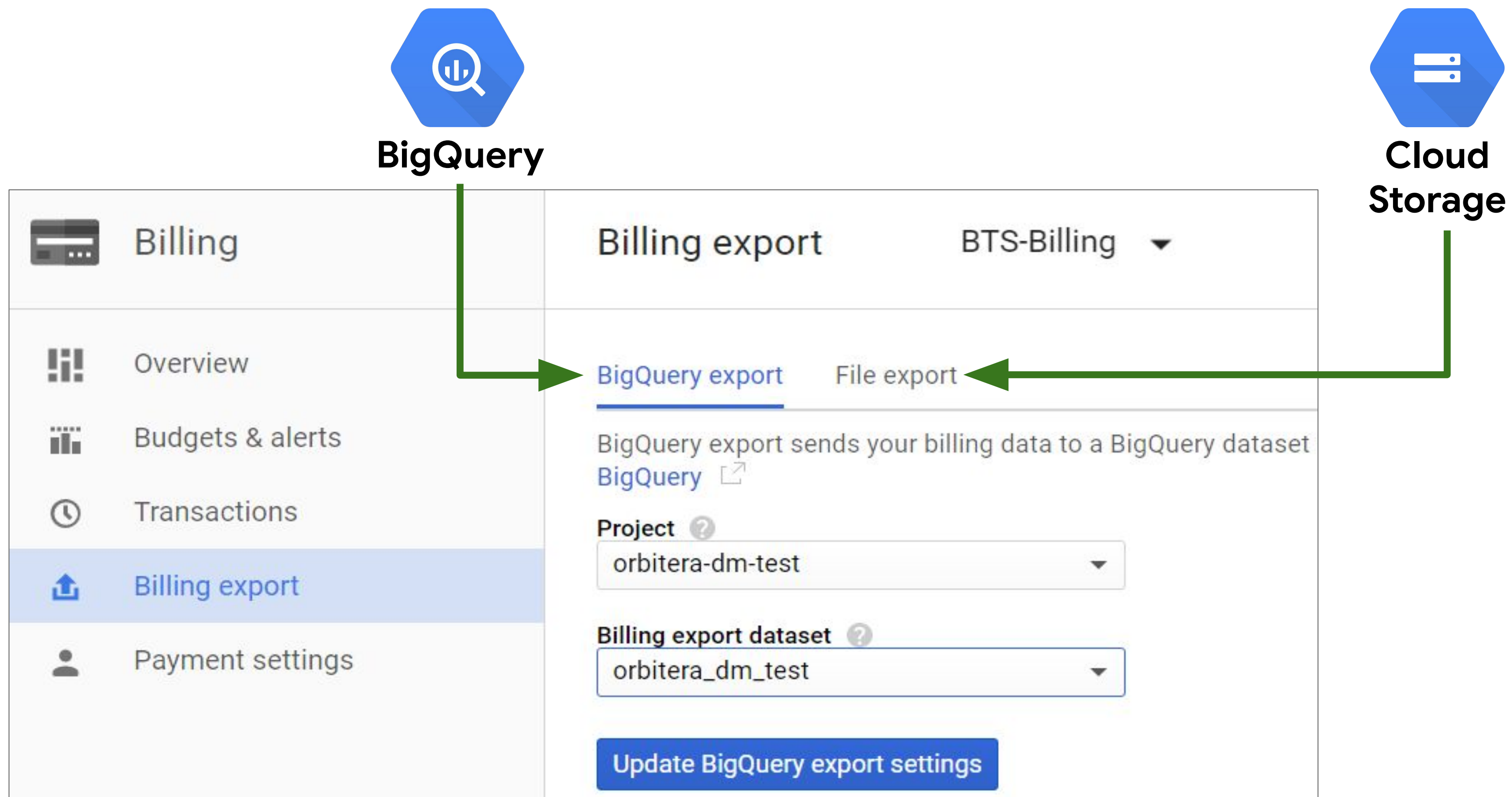
+ CREATE BUDGET

DELETE

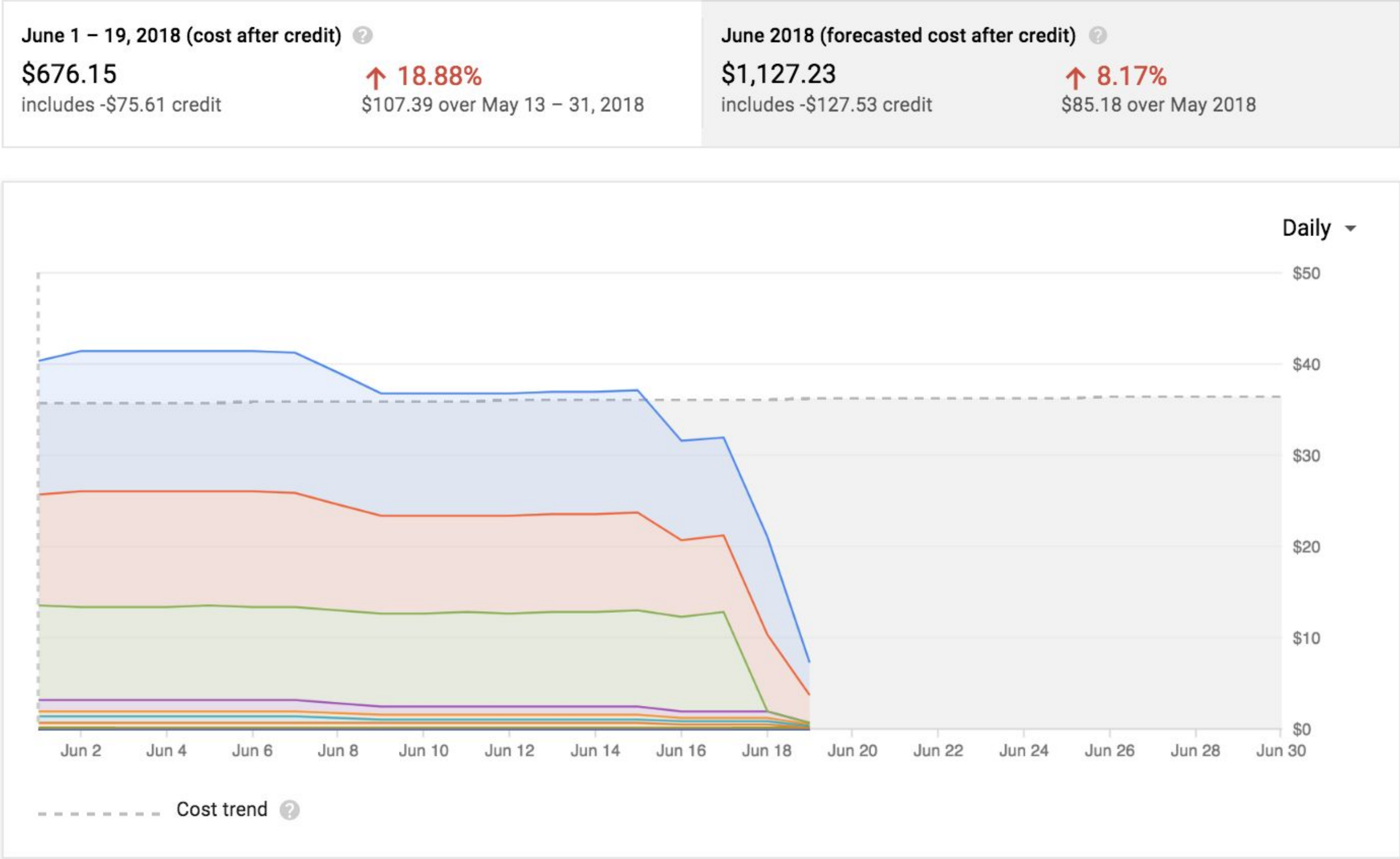
Budgets track expenses within a Google Cloud Platform project or billing account. Your budget can be a specified amount or based on previous spend. You can set alerts to notify billing admins when a budget goes over a specified amount.

☐ Budget name ^	Budget type	Applies to	Trigger alerts at	Spend and budget amount
☐ Trial budget 1	Specified amount	This billing account	50%, 90%, and 100%	\$86.34 / \$75.00

Billing export allows you to store detailed billing information



Reports is a visual tool to monitor expenditure



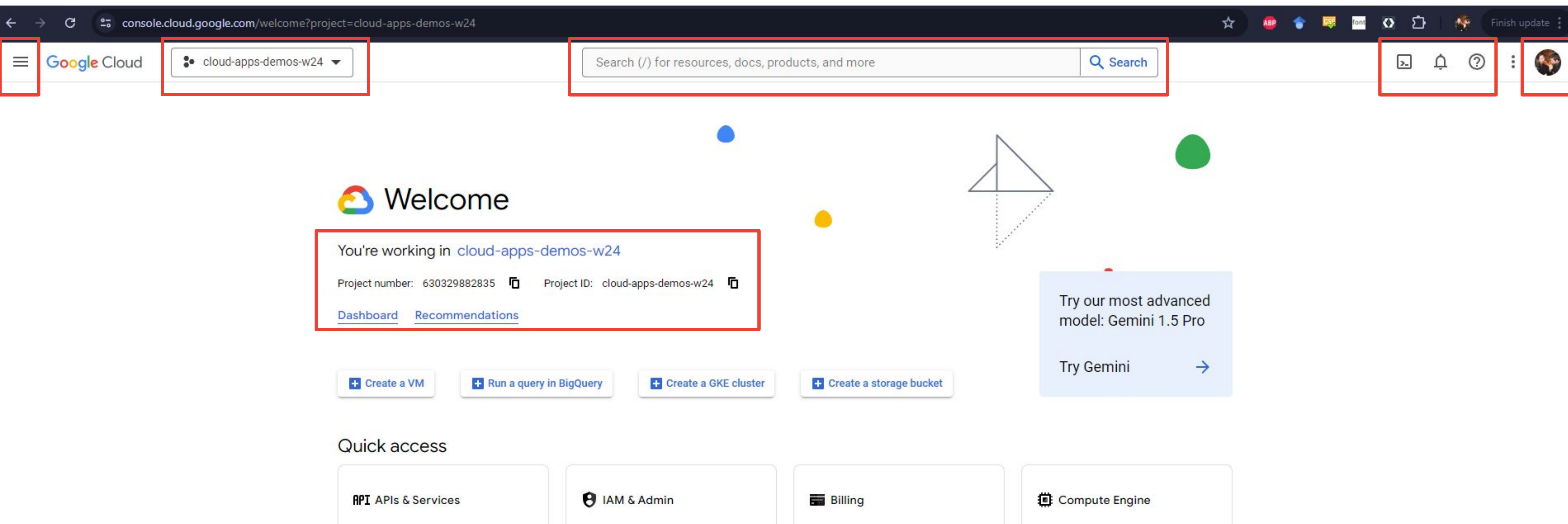
PROJECT	PROJECT ID	COST BEFORE CREDIT	CREDIT	COST AFTER CREDIT
● My Project 194	my-project-194-1378	\$287.07	-\$36.11	\$250.96
● My Project 301	my-project-301-1492	\$233.12	-\$31.00	\$202.12
● My Project 315	my-project-315-9812	\$175.00	\$0.00	\$175.00

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The interface (Google Cloud)

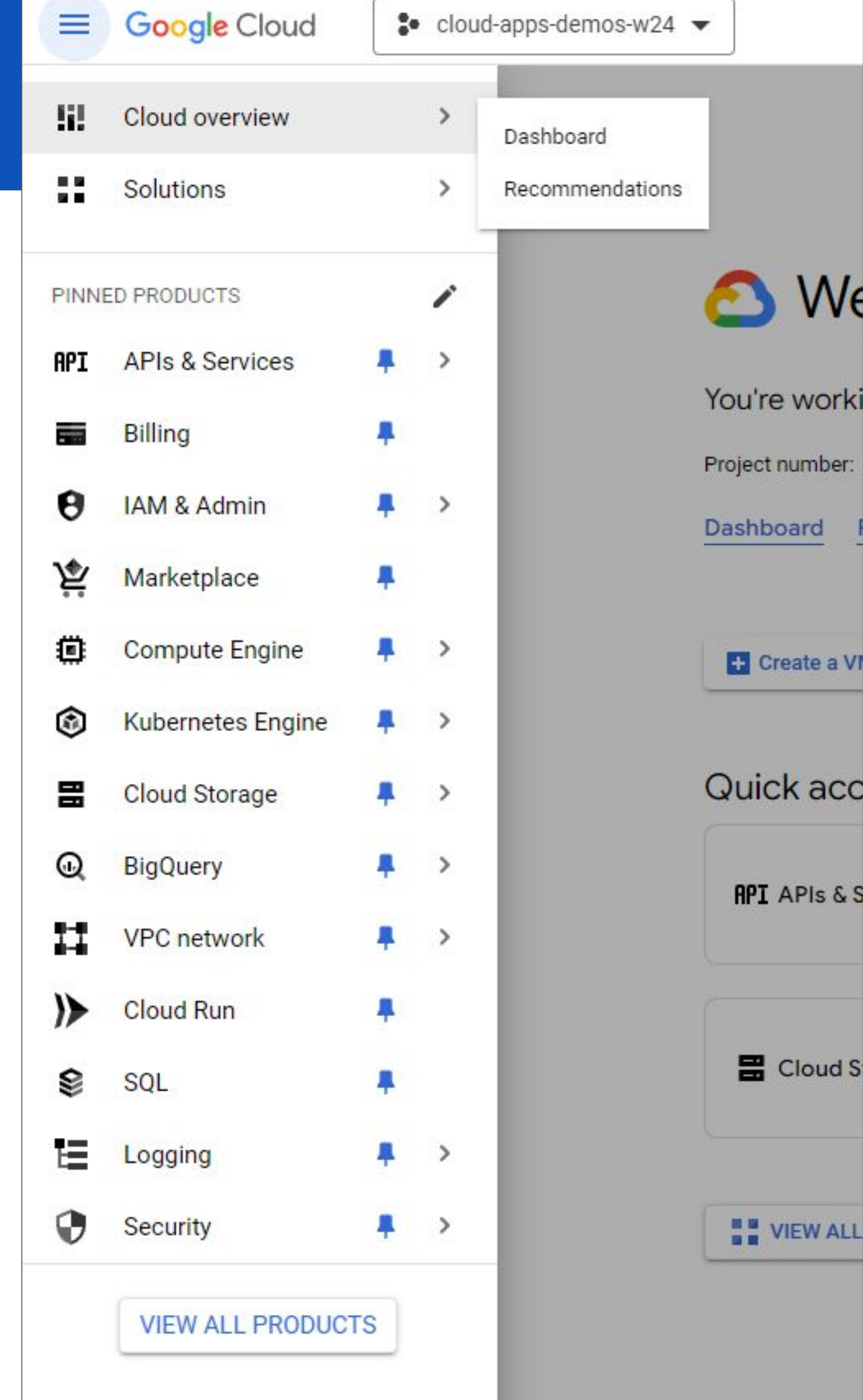


Google Cloud Interface (console.cloud.google.com)



With Google Cloud credits:

- **All** products are available to you!
- (and I can get you more *when* you run out)
 - NEVER PUT IN YOUR CREDIT CARD (AGAIN)



First...

Let's talk about Projects in Google Cloud

Every Google Cloud service you use is associated with a project

- Enable services and APIs.
- Enable billing.
- Manage permissions and credentials.
- Track resource and quota usage.
- Programmatically manage your projects in Google Cloud.



Projects have three identifying attributes

Project ID	Project name	Project number
Globally unique	Need not be unique	Globally unique
Assigned by Google Cloud but mutable during creation	Chosen by you	Assigned by Google Cloud
Immutable after creation	Mutable	Immutable

Creating a project

The screenshot displays the Google Cloud Platform (GCP) dashboard interface. At the top, a blue navigation bar contains the 'Google Cloud Platform' logo, a 'Project' dropdown menu (highlighted with an orange box), a search bar, and several utility icons (notifications, help, etc.). Below the navigation bar, the main content area is divided into three columns. The left column features three cards: 'Project info' (showing project name, ID, and number), 'Resources' (stating no resources are present), and 'Trace' (stating no trace data is available). The middle column displays the 'APIs' card, which includes a line chart of 'Requests (requests/sec)' over time, with a current value of 0.033. The right column contains three cards: 'Google Cloud Platform status' (reporting all services normal), 'Billing' (showing estimated charges of USD \$0.00), and 'Error Reporting' (indicating no errors are present). Each card includes a 'Go to' link for further action.

Google Cloud Platform Project

DASHBOARD **ACTIVITY** [CUSTOMIZE](#)

Project info

Project name
Project

Project ID
nth-skyline-247917

Project number
467846512270

→ Go to project settings

Resources

This project has no resources

Trace

No trace data from the past 7 days

APIs

Requests (requests/sec)

0.035
0.034
0.033
0.032
0.031

7:15 7:30 7:45 8 AM

Requests: 0.033

→ Go to APIs overview

Google Cloud Platform status

All services normal

→ Go to Cloud status dashboard

Billing

Estimated charges USD \$0.00
For the billing period Jul 1 – 30, 2019

→ View detailed charges

Error Reporting

No sign of any errors. Have you set up Error Reporting?

→ Learn how to set up Error Reporting

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Creating a project

The image shows the Google Cloud Platform 'New Project' dialog. A green arrow points from the 'NEW PROJECT' button in the top navigation bar to the dialog. On the left, a sidebar shows 'Select from NO ORGANIZATION' and a search bar. The dialog itself has a blue header with the Google Cloud Platform logo and a search bar. The main content area is titled 'New Project' and contains several fields: 'Project name *' with the value 'Project Example', 'Project ID' with the value 'project-example-248312' and an 'EDIT' button, 'Billing account *' with the value 'My Billing Account', 'Organization' (empty), and 'Location *' with a 'BROWSE' button. At the bottom are 'CREATE' and 'CANCEL' buttons. Green arrows point to each of these fields and buttons from the left.

Select from NO ORGANIZATION

NEW PROJECT

Search projects and folders

RECENT ALL

Google Cloud Platform

New Project

Project name *
Project Example

Project ID: project-example-248312. It cannot be changed later. EDIT

Billing account *
My Billing Account

Any charges for this project will be billed to the account you select here.

Organization

This project will be attached to

Location *
BROWSE

Parent organization or folder

CREATE CANCEL

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Deleting a project

Often you may be spinning up a project for class with the intention of deleting it when you're done

- For example, codelabs that require App Engine...



One other thing

At some point, you may receive a message that you have **too many projects and can't create another**

- At this point, **you** need to put in a request with Google to increase your capacity
- (I don't have control over your account!)

Another point re billing/projects

Each project is tied to a billing account

Meaning, if billing account has \$0, then project is inactive

- Can't be used
- Or edited
- Or any source code downloaded for storage
 - So back up anything critical often!
 - Examples of critical things to backup?

Eventually, projects with no billing account are deleted and **not recoverable**

Live Demo: Assignment 1

- Create a Google Cloud project
- Create an Ubuntu VM
- Install Apache
- Open the web server in a browser

Live Demo: Assignment 1

First, a look at the free tiers: <https://cloud.google.com/free>

Compute Engine

- 1 non-preemptible `e2-micro` VM instance per month in one of the following US regions:
 - Oregon: `us-west1`
 - Iowa: `us-central1`
 - South Carolina: `us-east1`
- 30 GB-months standard persistent disk
- 1 GB of outbound data transfer from North America to all region destinations (excluding China and Australia) per month

Your Free Tier `e2-micro` instance limit is by time, not by instance. Each month, eligible use of all of your `e2-micro` instances is free until you have used a number of hours equal to the total hours in the current month. Usage calculations are combined across the supported [regions](#).

Compute Engine free tier does not charge for an external IP address.

GPUs and TPUs are not included in the Free Tier offer. You are always charged for GPUs and TPUs that you add to VM instances.

[Learn more](#)

Live Demo: Assignment 1

Google Cloud

cloud-apps-demos-w24

Search (/) for resources, docs, products, and more

Search

1

?

Welcome

You're working in cloud-apps-demos-w24

Project number: 630329882835 Project ID: cloud-apps-demos-w24

[Dashboard](#) [Recommendations](#)

+ Create a VM

+ Run a query in BigQuery

+ Create a GKE cluster

+ Create a storage bucket

Quick access

 APIs & Services

 IAM & Admin

 Billing

 Compute Engine

 Cloud Storage

 BigQuery

 VPC network

 Kubernetes Engine

 VIEW ALL PRODUCTS

Try our most advanced model: Gemini 1.5 Pro

Try Gemini →

Yong Zhuang

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Welcome

Active project

You're working in [cloud-apps-demos-w24](#)

Project number: 630329882835

Project ID: cloud-apps-demos-w24

[Dashboard](#)

[Recommendations](#)

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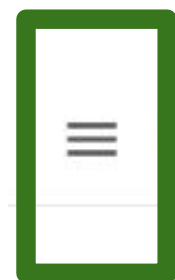
Cloud Storage

BigQuery

VPC network

Kubernetes Engine

VIEW ALL PRODUCTS



Google Cloud

cloud-apps-demos-w24

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1



You're working in [cloud-apps-demos-w24](#)

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[Dashboard](#)

[Recommendations](#)

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Try Gemini



Quick access

APIs & Services

IAM & Admin

Billing

Compute Engine

Cloud Storage

BigQuery

VPC network

Kubernetes Engine

VIEW ALL PRODUCTS

Where to find things



Live Demo: Create Your First VM

Google Cloud

cloud-apps-demos-w24

compute engine

Search

1

Create an instance

EQUIVALENT CODE

New VM instance

Create a single VM instance from scratch

New VM instance from template

Create a single VM instance from an existing template

New VM instance from machine image

Create a single VM instance from an existing machine image

Marketplace

Deploy a ready-to-go solution onto a VM instance

Name *

instance-20240528-212102

MANAGE TAGS AND LABELS

Region *

us-central1 (Iowa)

Region is permanent

Zone *

us-central1-a

Zone is permanent

Machine configuration

General purpose

Compute optimized

Memory optimized

Storage optimized

NEW

GPUs

Machine types for common workloads, optimized for cost and flexibility

Series	Description	vCPUs	Memory	Platform
☐ N4	Flexible & cost-optimized	2 - 80	4 - 640 GB	Intel Emerald Rapids
☐ C3	Consistently high performance	4 - 176	8 - 1,408 GB	Intel Sapphire Rapids
☐ C3D	Consistently high performance	4 - 360	8 - 2,880 GB	AMD Genoa
☒ E2	Low cost, day-to-day computing	0.25 - 32	1 - 128 GB	Based on availability
☐ N2	Balanced price & performance	2 - 128	2 - 864 GB	Intel Cascade and Ice Lake
☐ N2D	Balanced price & performance	2 - 224	2 - 896 GB	AMD EPYC
☐ T2A	Scale-out workloads	1 - 48	4 - 192 GB	Ampere Altra Arm
☐ T2D	Scale-out workloads	1 - 60	4 - 240 GB	AMD EPYC Milan
☐ N1	Balanced price & performance	0.25 - 96	0.6 - 624 GB	Intel Skylake

Machine type

Choose a machine type with preset amounts of vCPUs and memory that suit most workloads. Or, you can create a custom machine for your workload's particular needs. [Learn more](#)

PRESET

CUSTOM

e2-medium (2 vCPU, 1 core, 4 GB memory)

Monthly estimate

\$25.46

That's about \$0.03 hourly

Pay for what you use: no upfront costs and per second billing

Item	Monthly estimate
2 vCPU + 4 GB memory	\$24.46
10 GB balanced persistent disk	\$1.00
Total	\$25.46

[Compute Engine pricing](#)

[LESS](#)

CREATE

CANCEL

EQUIVALENT CODE

Fill out with free tier reqs!

Yong Zhuang

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Then, Install Apache

Typically the servers will be running Linux

- Much cheaper than Windows Server
- Get comfortable with Linux commands if you aren't!
 - <https://www.geeksforgeeks.org/linux-commands-cheat-sheet/>

(I'll be running them with Ubuntu mainly, but any install is valid as long as you know how to use it)

- AND SECURE IT
 -why?

And then, open ports in the firewalls

More than 1 firewall?

Inner firewall → your virtual machine

- YOUR RESPONSIBILITY TO MANAGE

Outer firewall → VPC network

- Provider's responsibility (though you specify ports)

MAKE SURE YOU OPEN UP SSH AS WELL (on the VM only)

- If you don't, you will lose the ability to remote into your machine
- Recovery instructions: <https://www.youtube.com/watch?v=8cKnIkYYsDQ>

What's the difference?

Internal IP	External IP
10.142.0.2 (nic0)	35.231.243.61 (nic0)

HOMEWORK

Homework posted -- accounts setup and web server setup